The O(1)-State Organism: Seven Measured Foundations for Continuously Living Sequence Models

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Draft v0.2, August 5, 2026

Every number in this paper is committed in the public repository ***github.com/DT-Foss/o1-state*** *with its generating script and raw JSON; the prediction ledger (****analysis/PREDICTIONS.md****) was written before the data it scores. This draft is itself part of the public disclosure.*

Abstract

Contemporary sequence models are batch artifacts: trained on a frozen corpus, frozen at deployment, and structurally incapable of one thing no scaling law addresses — **they never experience their own future**. We present measured foundations for the alternative: a bounded-state organism that consumes an unbounded stream at constant memory, decides *from its own prediction error* when to learn, what to remember, when to consult an external index, and when to sleep — and whose complete living state is a small, portable, forkable asset. Three measurements carry the case. **(i)** In a pre-registered 40-hour experiment (909.7M streamed tokens, 16/16 integrity checks), the organism's surprise gate did not merely approach full-gradient training at 25.2% of the gradient tokens — it **beat** it (improvement ratio 1.0091), crossing the prediction's own registered "embarrassment threshold." A three-point width curve then *falsified* the two-point reading of that result: the ratio peaks at d=256 (0.9729, 0.9953, 0.9777) rather than rising with width. What survives is sharper — per gradient token actually spent, the widest arm is the best of the three (0.3547 vs. 0.2969/0.2937). **(ii)** A 1,713,673-parameter organism has now streamed **7.05 billion tokens in one continuous life** at a process memory that never left 0.69–0.83 GB, 0.87B→7.05B of it in a single OS process over roughly twelve days; its one stall self-healed from its own atomic checkpoint. **(iii)** Two live machines on a real network, source never paused: at the one equal chunk count the held-out delta between an ARM and an x86 replica is **0.0, exact**. We further measure: an exactness license swept over four operators and six (chunk, overlap) points — layout decoupling exact to 0.0, gradient exactness gated by warmup *overlap* rather than chunk length; a two-timescale law from three independent experiments; keyed holographic recall through silence with a movable persistence knee; collective memory (one organism's stored surprises cut another's shock-forgetting to 0.67×); and family-generic transfer of the entire operating mode to a Mamba/S6 parametrization (0.98×). Forty-two predictions were registered before their data; the falsifications — a replay-cost law, a rank hypothesis, two migration thresholds — are reported at the same strength as the confirmations, with the numbers that killed them. We argue these primitives compose into a **reality prediction engine**: a system that lives on streams of the present, deposits predictions about its own future at every horizon, and learns from their arrival — an axis of capability orthogonal to parameter scaling, and one on which batch models cannot compete by construction.

1 The blind spot

A language model trained on a fixed corpus optimizes next-token prediction on the *past*. At deployment its weights are dead; its context window is a purchased illusion of memory; a session's experience evaporates. Three structural consequences follow. (i) The model cannot *verify* its own predictions — verification requires living until the predicted moment arrives. (ii) It cannot *accumulate* — its state is either bounded-and-overwritten or unbounded-and-unaffordable. (iii) It cannot *move* — the serving process is welded to the hardware that holds its KV cache.

This paper measures the alternative at small scale and full methodological rigor. The claim is not that a 128-dimensional organism rivals a frontier model at language; it does not. The claim is that the **operating layer** every persistent deployed model will need — when to learn, what to keep, how to move, how to sleep, how to verify — is measurable *now*, that its laws are architecture-generic, and that they are already surprising: the learner's own prediction error turns out to be not merely a sufficient control signal but a *better* teacher than learning from everything.

2 The substrate (prior work, compressed)

The organism runs on GSSM-Selective, a gated bounded recurrence $z_t = \gamma_t z_{t-1} + a_t$ in which Mamba/S6 [3], S5 [8], and LRU [6] are switch restrictions of one affine prefix-scan operator (family reduction measured at ∼1e-15; repository §1). Three prior structural results carry the present work: **(a)** the operator is shift-equivariant with a ∼5–8-token contraction receptive field, giving perplexity that *improves* to 524,288× the training length (§8) and a billion streamed tokens at flat RSS; **(b)** truncated-BPTT with a detached carried state reproduces full-window gradients exactly — streaming *training* is exact, not approximate, under a condition this draft measures precisely (§6); **(c)** gated readouts have a sharp capacity cliff at load ≈ 1 (slope 1.32 vs. 0.57 linear), which *forces* the two-system split between a bounded live state and a growable external index.

3 Seven foundations

Each is stated over the operator class and anchored by committed measurements. (Full formal statements: `FOUNDATIONS.md`.)

F1 — The surprise calculus. The learner's own prediction error suffices as the universal control signal: when to learn (gate), what to remember (spike-selected spans), when to consult (retrieval triggers), when and how much to sleep (dividend-monitored replay), how curious to be. *Anchor:* the 40h result of §4 and its width curve (§5). *Extension (disclosed, first experiment registered):* a ladder of horizons — deposit predictions about the stream's own future, score them on arrival, learn at every timescale.

F2 — The exactness license. Bounded contraction makes detach-carry streaming exact and decouples training layout from deployment layout. Swept over four operators × six (chunk, overlap) points, the license holds with a precisely located boundary: layout decoupling is exact to 0.0 everywhere; gradient exactness is bought by warmup *overlap*, not by chunk length (§6).

F3 — Phase–magnitude separation. In complex bound states, content (phase) is written and never evolves (zero-drive invariance $|\Delta\varphi| \approx$ 1e-8, machine precision); persistence (magnitude) decays. Real fillers are *double agents* — they pollute the phase (removable exactly by an eval-time clamp:

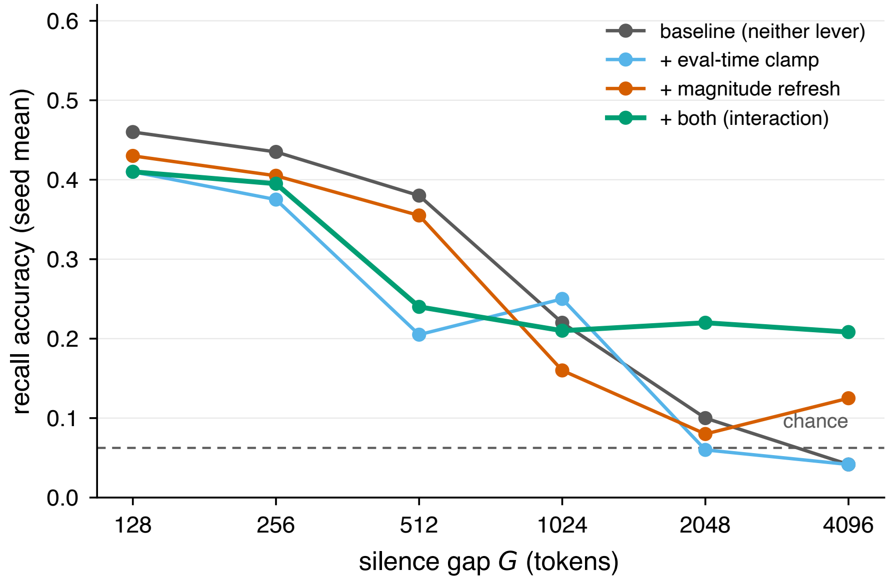

Figure 1: The persistence knee moves with the eval-time levers. Each lever alone decays toward chance beyond $G{\approx}512$–1024; applying both together holds recall at more than 3× chance out to $G{=}4096$ — an interaction, not a sum. Source: `results/holo_clamp_refresh.json`.

drift 0.0 rad) while *feeding* the magnitude (clamped, channels starve). The recall knee moved 32 → 256 → 512 → 2176 mean / 4096 end-of-range across theory-led interventions, the last requiring BOTH levers jointly — an interaction, not a sum. Knee *position* is seed-dependent; the intervention *effects* replicate.

F4 — The two-system law. The gated-readout cliff (rank-1-per-channel binding, $D_{\text{eff}} \approx 1.02$ measured against a closed-form bound) forces unbounded accumulation into an external index. Runtime consultation lifts recall far past state capacity: at 16 pairs, 0.150–0.158 unaided → 0.310–0.512 consulted, dose-dependent (a full gate roughly doubles the half gate) with the random-injection control at 0.100–0.143. Training *with* stochastic consultation makes the read near-perfect at low load — 0.995–0.998 at 2 pairs, against 0.530–0.565 unaided — and still lifts 16 pairs to 0.487–0.505. Held-out-key controls kill the lookup-table explanation: the binding generalizes to never-trained keys.

F5 — Family-generic operating modes. The full POS recipe run on a scan-parameter-matched Mamba/S6 configuration transfers at 0.98× (three seeds, two CPU architectures) — and the house operator beats S6 head-to-head (+0.13 to +0.16 nats) at parity. The calculus attaches to the operator class, not to one network.

F6 — Train short, deploy unbounded. One shift-equivariance symmetry on two axes: sequence length (perplexity improving to 524,288×; corpus-as-iterator to 10^9 tokens and beyond) and silence (keyed bindings held through gaps 128× the training horizon). Its honest boundary is measured twice: multi-chunk training is structurally gradient-blind to the write — the cure is to train inside the boundary and let exactness carry the skill out.

Table 1: The 40-hour surprise-gated run vs. full-gradient training.

arm	held-out 8.6588 →	improvement	gradient tokens
A2 full-gradient	4.7782	3.8806	100%
A3 surprise-gated	**4.7430**	**3.9158**	**25.17%**

F7 — The portable organism. The complete living system — weights, optimizer moments, carried state, gating windows, span store, stream position, RNG — is one atomic ∼53 MB artifact, and the organs couple through *kilobytes* (spans, reminders, deltas), not activations. Measured: local checkpoint→new-process→resume is bit-identical; **live mid-stream migration from Apple Silicon to x86 continues behaviorally identically to six decimals** (heldout 6.182391 == 6.182391, every gate decision matched — only the BLAS bit-digest differs, and it does not propagate); the same parity holds *between two live machines on a real network* (§10); a killed replica rejoining through a *shared* span store ends *better* than its never-killed twin (−0.014: the partner's outage-window spans are a gift); an outage spent sleeping beats one spent idle (+0.065).

4 The 40-hour experiment: the gate beats the firehose

One process, three from-scratch arms on one cloned C4 stream (seed 42), 40.0 hours, 909,676,544 tokens, no restarts, RSS $\leq$ 1.094 GB. A1: forward only. A2: backward on every chunk. A3: backward only when the chunk's own mean NLL clears the rolling 75th percentile of its own history — no oracle, no second model, no labels, no schedule.

Ratio 1.0091. The pre-registered prediction (P1) put the point estimate at 0.85 with an explicit embarrassment threshold: *"ratio > 1.0 sustained would be a bigger result than the thesis itself."* It fired. The surprising part is not efficiency — it is that the 75% of chunks the gate skips are, in net, *worth skipping*: selective plasticity beats indiscriminate plasticity at equal data exposure. Every number re-derives from the raw 1.78M-chunk log (16/16 integrity checks).

The twin. At T+24h, A3's weights were copied into a fresh process with zero carried state. Registered: a warmup tax (+0.03–0.15 excess surprise, $\geq$ 1.5× over-gating, hours to converge). Measured: excess 0.0029, post-fork gate rates 0.2521 vs. 0.2522, convergence one chunk after the fork, and the twin finished *ahead* (4.7365). Restarting a living stream is free at the fast timescale — the third independent measurement of the two-timescale law (§9).

The honest negative. 1020 paired index-injection probes: both deltas negative (the index froze at its 20k cap by token 11M; replaying 5–11M-era spans into an 80×-older model disturbs), but injection hurts 3.2× *less* than random injection (−0.087 vs. −0.279; helped 0.35 vs. 0.10) — the mechanism carries information; the provenance was stale. Registered as falsified; the v2 (rolling spike threshold) is specified.

5 The width curve: a two-point law, falsified by its own third point

The natural next question is whether the gate's advantage survives width. P42 registered it as a scaling axis of the core result, with the point call stated in advance: the ratio at d=256 should land at or above the d=128 anchor, because selection ought to matter *more* where each gradient is four times as expensive. The same recipe (q=0.75, window 500, B=8, K=64, lr 3e-3, seed 42, from-scratch, streamed C4) was run with only the width changed, all three points read at a matched ∼50M-token anchor.

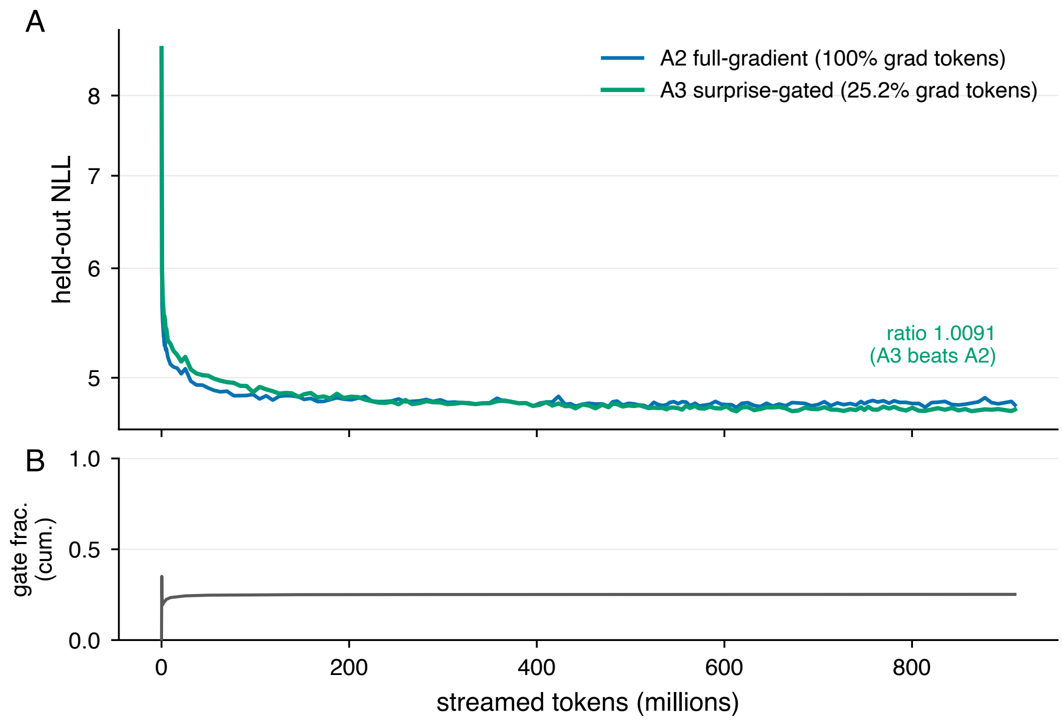

Figure 2: The 40-hour run: held-out loss for A2 (full-gradient) and A3 (surprise-gated), and A3's cumulative gate fraction, vs. streamed tokens. A3 tracks A2 while training on 25.2% of the gradient tokens and finishes ahead (ratio 1.0091). Source: `results/pos_metrics.jsonl`.

At two points the reading was clean and was reported as such: 0.9729 → 0.9953, the gate law *grows* with width. **The third point falsifies it.** The ratio does not rise monotonically; it peaks at d=256 and falls to 0.9777 at d=512. The two-point trend was an artifact of having only two points, and the third was run on purpose to find out.

What survives the falsification is sharper than what it replaced, and it turns on a dosing confound the artifact records. The three arms are not equally dosed: the d=512 gate fires at 19.84% where both narrower runs fire at 24.72%, so A3 there receives roughly 20% fewer gradient tokens. **Per gradient token actually spent, d=512 is the best of the three** — 0.3547 against 0.2969 and 0.2937. Selection did not get worse with width; it got cheaper. The ratio fell because the dose fell.

Two candidate explanations for the smaller dose were measured and both were refuted. The surprise distribution does *not* change shape with width: relative spread is 0.03813 / 0.03544 / 0.03404 and absolute standard deviation 0.163 / 0.164 / 0.161 across d=128/256/512. Nor does

Table 2: The gate-law width curve at a matched ∼50M-token anchor. The registered metric (improvement ratio) peaks at d=256; per gradient token spent, the widest arm is the best of the three. Sources: `results/gate_law_width_curve.json`, `results/gate_rate_width_probe.json`.

d_{model}	gate fires	A3 gated	A2 full	improvement ratio	improvement / M grad-token
128	24.72%	5.017554	4.916141	0.9729	0.2969
256	24.72%	5.053426	5.036284	**0.9953**	0.2937
512	19.84%	5.170144	5.089800	0.9777	**0.3547**

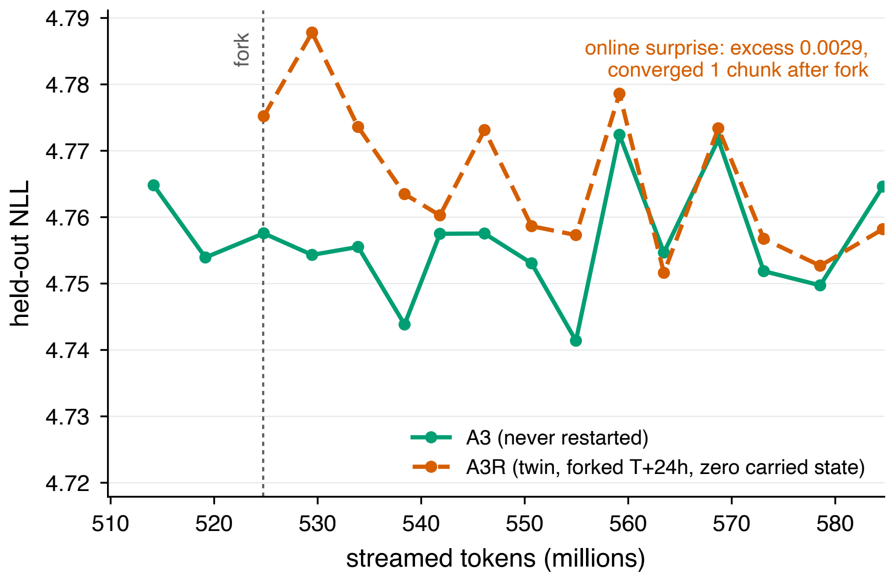

Figure 3: The twin, zoomed to the fork. A3R (forked at T+24h with zero carried state) opens ≈0.02 nats above A3, crosses into A3's own fluctuation band within a few evaluations, and finishes ahead (4.7365 vs. 4.7430). On the registered online-surprise criterion the restart is free: excess 0.0029, post-fork gate fractions 0.2521 vs. 0.2522, convergence one chunk after the fork. Source: `results/pos_metrics.jsonl`, `results/pos_summary.json`.

within-window drift separate the arms: the fraction of the last 200 window entries above the q75 threshold is 0.230 at *both* d=256 and d=512, while their gate rates are 0.220 and 0.185. What the data does locate is *when* the difference is born — the cumulative gate fraction at the first eval is already 0.2042 (d=256) against 0.1785 (d=512); over the remaining 46M tokens they drift up by +0.043 and +0.020, the separation growing 0.026 → 0.049. The rate is born during ignition, and the gap then widens rather than washing out. The per-chunk traces (pre-registered as P45) locate the birth further: the separation is established within the first 500 chunks, and across six cells (three widths × two seeds) the gate rate tracks the *depth* of the always-learn ignition descent — the model that exits ignition lowest gates least (r=0.77; n=6, not significant) — because the quantile window forms on the high descent trail. Width, however, does *not* set that depth: at the second seed the ordering reverses (d=256 exits lowest and gates least), and a same-seed repeat of one cell is bit-identical for 118 chunks, then forks at a quantile interpolation and lands at a different 2,000-chunk rate (0.163 vs. 0.191): early gate rates are a measured lottery. The 50M-scale evidence points the other way — the separation *grows* from 0.026 to 0.049 over the run, and d=128 and d=256 land on the same 24.7% cumulative rate, two widths finding one attractor. The repeats at the 50M scale close the question on both axes. An exact repeat of the d=512 run (11 days later, a different process, one mid-run stream reconnect) reproduces *all* 97,657 chunks identically on every recorded field — same-seed variance at 50M is zero, and the run-vs-run forks trace to measurement-cell co-load, not to the calculus. A second seed at d=512 lands at 0.1986 against 0.1984, with the full profile reproducing (improvement ratio 0.9758 vs. 0.9777; efficiency 0.3467 vs. 0.3547 per million gradient tokens). The early lottery is a transient, not a fate: lifetime cumulative rates converge to their width's attractor, and the dose structure of selection at scale is deterministic and plannable. (Seed-robustness is measured at d=512; the 24.7% side rests on two widths landing on one rate independently, one seed each at 50M.)

One control on the reading: A2 itself degrades with width at this fixed token budget (4.9161 →

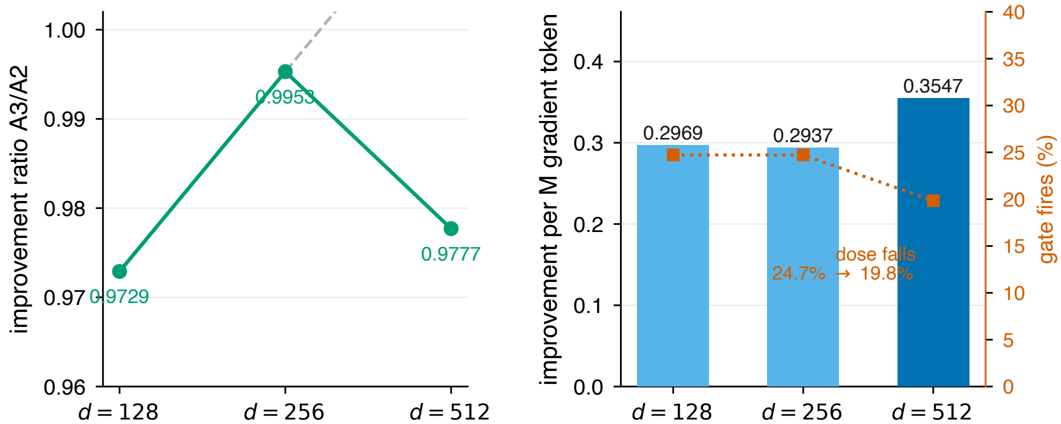

Figure 4: The gate-law width curve. *Left:* the registered metric. The two-point trend (dashed) does not survive the third point — the improvement ratio peaks at d=256. *Right:* what survives. Per gradient token spent, d=512 is the best of the three (0.3547), and the gate fires there at 19.8% rather than 24.7%: the ratio fell because the dose fell, not because selection weakened. Sources: `results/gate_law_width_curve.json`, `results/gate_rate_width_probe.json`.

5.0363 → 5.0898), exactly as expected — wider models need more tokens. The anchor measures selection efficiency, not model quality, and no claim about the latter is made from these three runs.

6 The exactness license, swept

F2 is the load-bearing license: if detach-carry streaming were merely approximate, every downstream claim about living training would inherit an uncontrolled error. v0.1 anchored it on one operator at one (chunk, overlap) point. A license with no measured failure mode is a claim, not a law, so the sweep runs four operator configurations — the selective scalar scan, the complex holographic scan, the same scan under the tanh-m readout, and a phase-off control — across six (chunk, overlap) points, including chunk sizes deliberately below the stated receptive field.

Two separate statements come out, and only one of them is unconditional. **Layout decoupling is exact to 0.0** — at every operator and every chunk size down to 16, with pure detach-carry (overlap 0), the chunked and full-sequence forward layouts agree to *max-abs-delta 0.0* exactly. Training layout and deployment layout are interchangeable, full stop. **Gradient exactness is gated by warmup overlap, not by chunk length.** With overlap 16 the gradient cosine is 1.000000000000 at relative error ∼5e-7; with overlap 0 the relative error runs 5e-3 to 2e-1 and the worst cosine over the whole sweep is 0.9762 (the complex holographic scan at chunk 16). Chunk length is not what buys exactness — a 128-token chunk with no overlap is worse than a 64-token chunk with overlap 16. The condition is that the carried state be re-warmed past the contraction receptive field before gradients are scored, which is precisely what the ∼5–8-token field predicts and what the ≥16-token overlaps deliver.

This supersedes the single-point anchor of v0.1 and states F2 with its boundary attached: the license is real, it covers every operator we run, and it has an operating condition that is measured rather than assumed.

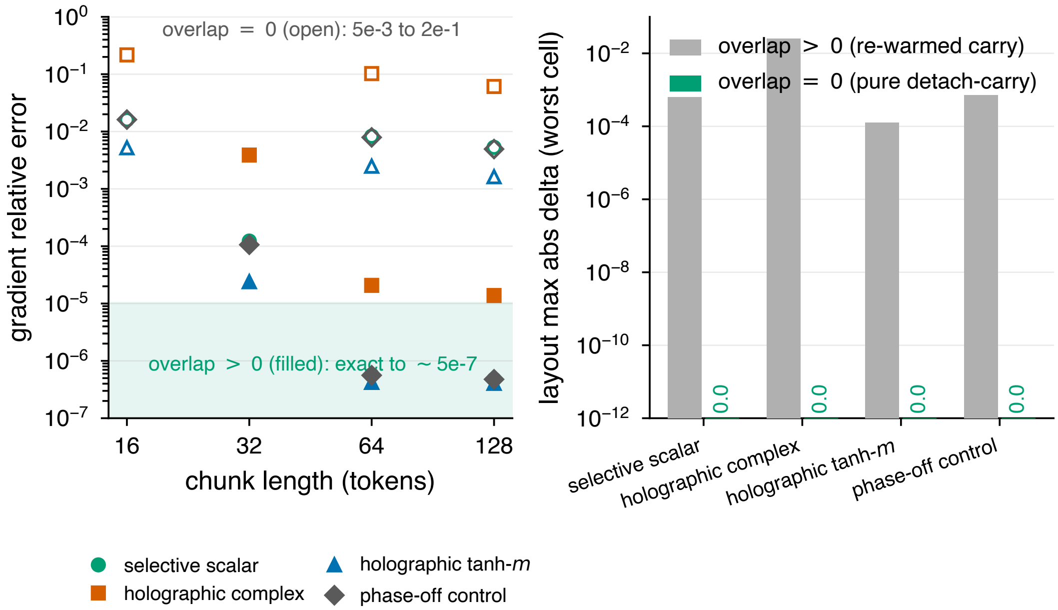

Figure 5: The exactness license, swept over four operators × six (chunk, overlap) points. *Left:* gradient relative error. Filled markers (overlap > 0) sit in the exact band at ∼5e-7 regardless of chunk length; open markers (overlap = 0) run 5e-3 to 2e-1. Exactness is bought by overlap, not by chunk size. *Right:* the layout decoupling is exactly 0.0 with pure detach-carry at every operator, including chunks below the receptive field. Source: `results/f2_equivalence_sweep.json`.

7 Seven billion tokens in one life

A 1,713,673-parameter organism has learned online through **7,053,926,400 streamed tokens in one continuous life** — no dataset stored, no epochs, no growing context — at a process memory that never left the band **0.69–0.83 GB**. The stretch from 0.87B to 7.05B tokens ran in a *single OS process* for roughly twelve days with zero interventions.

The one stall in that life is part of the result rather than an exception to it. On 2026-07-24 an upstream stream hang killed the process at 869,734,400 logged tokens; it resumed from its own atomic checkpoint at 869,683,200 — **51,200 tokens replayed, roughly three minutes of stream lost** — and continued for another six billion tokens without further intervention. The organism healed itself out of its own snapshot, which is exactly what F7 is for.

What this artifact does and does not claim. Over the logged stretch the loss EMA moves 4.25 → 4.169. At 1.7M parameters the model is at its capacity floor, and we state the consequence plainly: this measures the **constancy of resources over experience**, not continued learning. The cost of having lived through seven billion tokens is zero additional memory — that is the claim, and it is the one no batch model can make at any parameter count, because the quantity it holds constant is not a hyperparameter but a lifetime. (A further honesty note the artifact carries: the 0→512M stretch of this life predates the oldest retained log and is documented only by the run's status artifact and its start lines; the plotted series therefore begins at 512M tokens.)

Figure 6: The 7-billion-token life. Loss EMA (left axis) and process RSS (right axis) against tokens streamed through one O(1) state. Memory stays inside 0.69–0.83 GB across almost four orders of magnitude of experience; the marked stall is the single interruption, self-healed from the run's own atomic checkpoint. Sources: `results/lifetime_7b_series.json`, `results/lifetime_7b_curve.json`.

8 Length, re-anchored

The strongest length statement in this program is not flatness. Using the O(1) recurrent forward with chunked streaming (chunk 8192, overlap 128 $\gg$ the receptive field), a model trained at T=32 was evaluated on a single effective sequence of **16,777,216 tokens** — 524,288× **the training length** — at a peak RSS moving only 1.90 → 2.48 GB, and the perplexity *improves monotonically the whole way* (×1.000, ×0.896, ×0.886, ×0.825, ×0.803). The model is integrating causal context across the distance as a noise filter; it is not merely failing to crash.

The clean causal ablation says what the wall actually is. At 256× the training length, the position-free Selective arm sits at ×0.973 while the *identical* architecture with a sinusoidal positional encoding degrades to ×4.23 and the ungated Pure arm explodes to ×11.25. The only difference between the first two arms is the positional encoding. **The wall is the position term, not the length** — and both ingredients are necessary, since removing the gate breaks the bounded state on its own.

9 The two-timescale law of operation

Three independent experiments give the same law. (1) The twin (§4): zero carried state costs nothing. (2) A snapshot cross-matrix (weights at 359M tokens × states from 128M/240M/359M/zero/shuffled): every arm converges to the native trajectory within 4 chunks — 256 tokens, the receptive-field scale; a cold start is not worse than the carried state. (3) The beacon: a *stored bit* in a write-once-freeze carrier channel (measured at $\gamma = 0.9998$ in the gap, input gate 0.0133 there against 0.5652 at the beacon) survives a d64→d128 widening surgery at recall 1.000 and survives a weight swap across 2× training distance at recall 1.000 through 512 tokens of silence, against a zeroed-state control at chance (0.475–0.54) — by redundant encoding (the carrier's channel address can relocate; the read still lands).

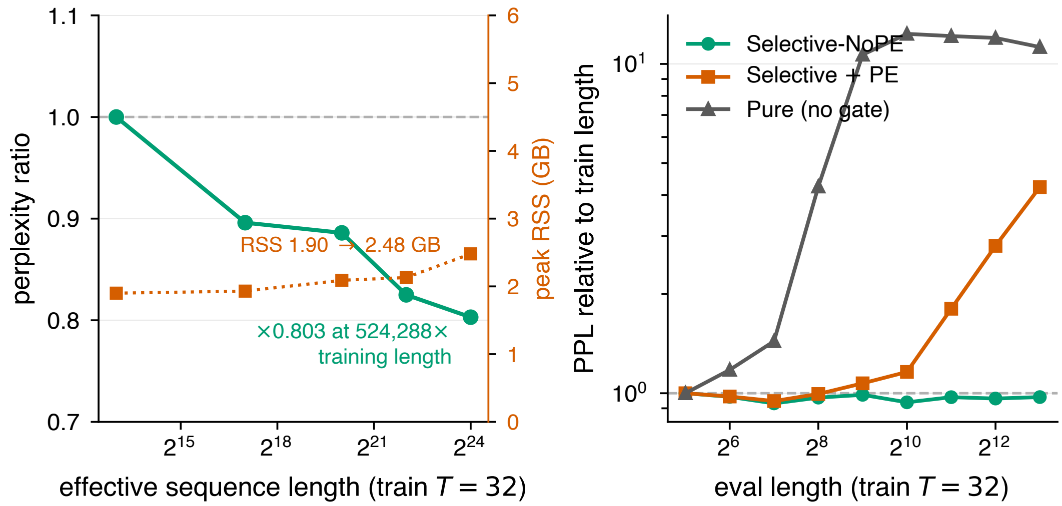

Figure 7: Length, re-anchored. *Left:* perplexity ratio and peak RSS against effective sequence length — ×0.803 at 524,288× the training length, improving monotonically, at 1.90→2.48 GB. *Right:* the control. Position-free Selective holds ×0.973 to 256×; the same architecture with a positional encoding degrades to ×4.23 and the ungated arm to ×11.25. Sources: `results/scale_to_a_million.json`, `results/length_extrap_v2_extreme.json`.

The law: the fast path forgets by design — so weight updates, model growth (function-preserving widening: surgery equivalence 6.7e-6, no transient, growth beats restart by +0.13 to +0.17 nats, 3× replicated), and migration are *safe live operations*; everything durable lives in slow carriers, weights, and the index — exactly the layer the organism's own calculus (F1, F4) manages. Operationally this is the answer to a question every persistent-model deployment will face: *what must be preserved across an update?* Measured: almost nothing — if the architecture separates its timescales.

10 Two live machines, one organism

v0.1 reported cross-ISA migration as a checkpoint-and-resume operation. The stronger staging runs it *live*: organism A streams continuously on an Apple Silicon machine and is **never paused**; organism B rises on an x86 server from A's snapshot transferred over real `scp`/`ssh` and chases A's stream position by deterministic replay. At the one chunk count where both organisms stood (560), the held-out delta is **0.0 — exact**: ARM 6.199778 against x86 6.199778, reproduced across two independent stagings.

The limit of the claim is structural and we state it as the artifact does. **Parity is defined only at equal chunk counts.** B on 16 server cores outruns A on the Mac and overtakes it; at B's 5480 chunks A had never reached that position (its maximum was 3000), so no comparison exists there — and a comparison at unequal stream positions would not be a parity check at all. Multi-cycle parity therefore requires rate-matched machines. The single-cycle number is the measured one, and it is exact.

Table 3: P39 (a)/(c) measured in situ at the production cadence (d=128, B=8, K=64 — the exact 40h configuration). Source: `results/p39_two_machine_scored.json`.

machine	cores	(a) gap ratio	threshold	(c) CPU ratio B/A	threshold
Mac (ARM)	10	7.085	< 3.0	2.219	< 1.0
beast (x86)	16	5.603	< 3.0	2.039	< 1.0

11 The replay-cost law: two falsifications and the mechanism behind them

P39 registered stop-free migration with four checks. Check (b) — B reaches state parity with A while A never pauses — passes at every cadence and on every machine tried, including the cross-ISA staging above. Checks (a) and (c) are **falsified as registered**, on two independent machines.

The obvious explanation was tested and refuted. If (c) failed because A and B contend for cores on one machine, then giving B its own cores among sixteen should fix it; it moves the ratio by 0.18, not by the 2× it would need. The raw CPU seconds locate the cause instead: **B spends 45.4 s replaying the chunks A produced with 22.3 s of CPU** — replay costs approximately **twice** live streaming per chunk, on both ISAs, contended or not. Check (c) as registered asks replay to run *faster* than production, and nothing in these two runs suggests it can. That is a property of catch-up replay, not of any machine, and it is the standing constraint on continuous replication: a replica cannot catch up on a machine no faster than its source.

A second, methodological lesson rides along. Earlier ratios for the same two checks were inflated by toy chunk weight — 17.935 and 3.789 at B=4/K=32, roughly halving to 7.085 and 2.219 once a chunk carries production work. The mechanism is arithmetic: snapshot size scales with the model, not with chunk weight, so raising chunk weight shrinks the snapshot cost measured in chunk slots (7.94 → 2.59 slots). The cadence hypothesis is therefore half confirmed and half falsified, and both halves stand: toy cadence really did inflate these ratios, and cadence was not the whole story — a residual survives that no cadence correction removes.

12 The organism is social and collective

Organism A streams a C4+code mix, storing its surprise spans. Organism B, facing a code shock, replays A's shared spans and forgets **0.67**× of what it forgets with only its own spans (0.155783 vs. 0.233120) — at better plasticity (0.542204 vs. 0.517594) — while token-shuffled A-spans do not beat B's own private spans (0.317260, against 0.375721 for no replay at all): the benefit is content, not regularization. This is replay as a continual-learning mechanism [10] with the selection made by the learner's own surprise rather than by a curator, and with no weight-space penalty [4] in the loop. Combined with F7: a fleet of small organisms sharing an index is a *measured* design, not an aspiration — including the striking corollary that a replica which dies and rejoins through the shared store can end *better* than one that never died.

13 Method as contribution: the adversarial ledger

Every experiment in this program was preceded by a numbered, committed prediction (P1–P42 at this writing); a scoring script audits result JSONs against the register; falsified predictions remain

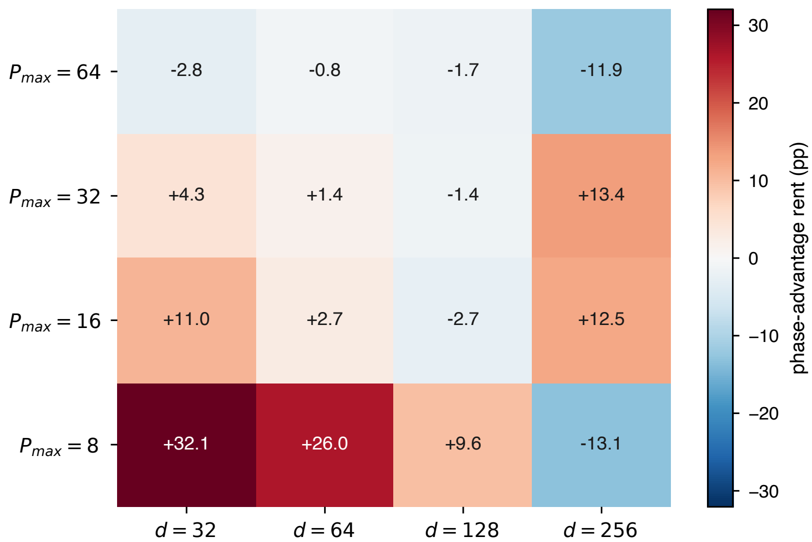

Figure 8: The 16-cell rent map: phase-advantage rent (percentage points) as a function of the recall load $P_{\max}$ and width d_{model}. The one-parameter rent law that motivated a single scalar prediction is falsified by this map's structure; the map explains, cell by cell, where the phase advantage on real text holds and where it does not. Source: `results/holo_rent_map.json`.

in the ledger with the numbers that killed them. The register is immutable in its P-numbers: a prediction is amended only before its data exists, with the reason on record, and never after.

Of the scored predictions, roughly one third were falsified or partial — including the program's own favorite hypotheses. Four are load-bearing enough to name with their killing numbers.

- **The width law (P42, §5).** "The gate law grows with width" held at two points and died at the third: 0.9729, 0.9953, 0.9777. The third point was run to test the trend, not to confirm it.
- **Replay cost (P39a, P39c, §11).** Registered thresholds <3.0 and <1.0; measured 7.085/5.603 and 2.219/2.039 on two machines, with the shared-core explanation refuted and the 2× replay-cost mechanism located.
- **Relational rank (P34).** Registered: the phase lifts per-channel binding rank to ≥2× the scalar's. Measured on a 4-arm × 5-load × 4-seed sweep — the attention control solves every load (test recall 0.9888 to 0.9997, so the harness is sound and the task solvable), while both state arms sit at chance from K=4 on (chance 0.015625). The registered relational criterion had exactly one eligible cell, at K=2, where the ratio is 0.371 — the opposite sign of the prediction — and the cliff ratio is 1.0 against a registered ≥2. The rank check itself is reported as *insufficient data* rather than as a pass or a fail: no load clears 3× chance for the phase arm with the required two ignited seeds, so no cell was eligible to score. The dominating phenomenon is a different one, and it is a finding: phase ignition collapses with load (3/4 ignited seeds at K=2, 0/4 by K=16). Whatever the phase might pay at high load, training reliability dies first.
- **The one-parameter rent law (P32).** Falsified *and then explained* by a 16-cell rent map measured the same night (Figure 8).

Two method-level findings emerged from the discipline itself: **training-dynamics sensitivity to floating-point reduction order near ignition boundaries** (thread count changes *whether* a model ignites, not just speed — 6/6 seeds ignite in one regime, 0/4 in another), and the resulting rule that instruments must reproduce their reference regime. A third, from §11: an instrument must reproduce its reference *cadence* as well, or it measures overhead arithmetic rather than the property under test. We offer the ledger pattern itself as a transferable practice: a vision paper whose every claim is either a committed measurement or an explicitly open registered question.

14 The reality prediction engine (the road this opens)

The foundations compose into something none of them states alone. A system that (i) lives indefinitely on present streams at constant memory (F2, F6) — measured to seven billion tokens (§7), (ii) controls all plasticity from its own prediction error (F1) — in the spirit of temporal-difference learning's use of prediction error as a teaching signal [9] and predictive coding's use of prediction error as the currency of cortical inference [7], (iii) can *deposit* a prediction about its own future in a persistence mechanism built for exactly that (F3's carrier; measured to survive silence, surgery, and weight updates) — the world-model commitment articulated for autonomous machine intelligence [5], (iv) scores the deposit when the future arrives — a loop only a *living* system can close, because a batch model never experiences its own future — and (v) exists as a portable, forkable, shardable, collectively- remembering asset (F7, §10): this is a **reality prediction engine**. Its scaling axis is not parameters but *scope* — more streams, more horizons, more organisms sharing more index. The first horizon-ladder experiment (early-warning under distribution shift; the long-horizon head as gating teacher) is registered (P37) and running at this draft's commit.

15 Limitations, stated plainly

Scale: $d_{\text{model}} \leq 512$, WT-2-derived vocabularies; no claim of frontier language quality is made or implied, and the width curve (§5) measures selection efficiency at a fixed token budget, not model quality. The lifetime organism's constancy claim is about resources, not about learning: at 1.7M parameters it is at its capacity floor. Cross-ISA parity is measured at equal chunk counts only, and multi-cycle parity needs rate-matched machines (§10). Catch-up replay costs $\sim 2\times$ live streaming, which bounds continuous replication (§11). Gradient exactness requires a warmup overlap past the receptive field; with zero overlap the relative error reaches 2e-1 (§6). The phase mechanism's ignition is fragile (seed-, load-, and reduction-order-dependent) — a measured fact that bounds current practice and is itself a finding. The injection result is provenance-limited (stale index era). Wall-clock throughput numbers are never claimed (host contamination); all axes are token-based. The rank-1 anchor's generating script is lost (artifact-backed, reconstruction leads documented) — reported as reproduction debt.

16 Invitation

Everything here reruns from the repository: the harnesses, the exact configurations, the raw JSONs, the ledger, and the scoring script. The organism that produced §4 can be resumed, forked, migrated, or extended by anyone — that is what F7 is for; the organism of §7 was still streaming when this draft was compiled.

Related work: predictive coding and JEPA (the vision lineage) [7, 5], TD-learning (multi-horizon error) [9], replay and weight-space penalties in continual learning [10, 4], zoology/MQAR (recall *instruments) [1], Mamba/S5/LRU (the operator family) [3, 8, 6], VM live migration (the systems analogy) [2].*

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